

XGBoost Coffee Quality Prediction

Comprehensive Analysis with Multiple Modeling Approaches

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Abstract

This report analyzes the Coffee Quality Institute's Arabica coffee dataset using XGBoost regression to predict Cupper Points (expert quality scores). The analysis compares multiple feature sets (Sensory, Extended, Full), evaluates baseline models, and documents both successful approaches and those found less effective. **Key finding:** sensory features dominate prediction, with XGBoost providing modest improvements over Linear Regression that are **not statistically significant** ($p = 0.725$).

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1 Introduction

This report analyzes coffee quality prediction using XGBoost regression with multiple modeling approaches.

Successful Approaches:

- Full model comparison (Sensory, Extended, Full feature sets vs baselines)
- Enhanced evaluation metrics (MAE, MAPE, Adjusted R^2 , Max Error)
- Learning Curves for training dynamics visualization
- Explainable AI (PFI, PDP, Break Down)

Approaches Tested but Found Less Effective:

- Lambda (L2) and Min_Child_Weight regularization tuning
- Bayesian Optimization vs Grid Search comparison

Table 1: Analysis Configuration

Parameter	Value
Random Seed	42
CV Folds	10
Train/Test Split	80/20
Early Stopping	50
Max Rounds	2000

2 Data Loading and Preprocessing

Dataset: 1311 observations, 41 columns after preprocessing.

3 Exploratory Data Analysis

3.1 Target Variable Analysis

Table 2: Target Variable (Cupper.Points) Summary

Statistic	Value
Mean	7.4979
Std Dev	0.4746
Median	7.5000
Min	0.0000
Max	10.0000
IQR	0.5000

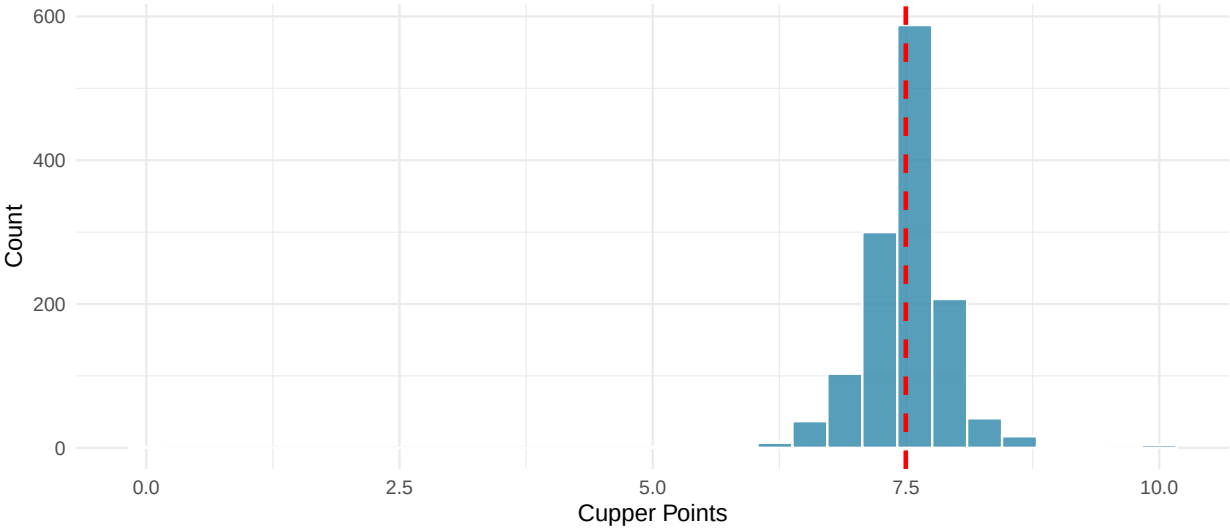


Figure 1: Distribution of Cupper Points. Red dashed line indicates mean.

3.2 Correlation Analysis

Table 3: Feature Correlations with Cupper.Points

Feature	Correlation
Flavor	0.7971
Aftertaste	0.7878
Balance	0.7415
Acidity	0.7004
Aroma	0.6934
Body	0.6711
Uniformity	0.3585
Clean.Cup	0.3570
Sweetness	0.3001

Feature	Correlation
Quakers	0.0119
Category.One.Defects	-0.0527
Moisture	-0.1792

Critical Note on Multicollinearity: The correlation matrix reveals high correlations among sensory features (e.g., Flavor-Aftertaste: 0.90, Flavor-Acidity: 0.82). This multicollinearity affects coefficient interpretation in linear regression and may cause PFI to underestimate importance of correlated features due to information redundancy.

4 Feature Engineering and Data Preparation

Table 4: Feature Set Definitions

Feature.Set	N.Features	Description
Sensory	12	Core sensory quality indicators only
Extended	16	Sensory + metadata (processing, variety, altitude)
Full	23	All features including origin and logistics

Final dataset: 1310 observations. **Train:** 1048 (80%). **Test:** 262 (20%).

5 Baseline Models

Table 5: Top Linear Regression Coefficients

Feature	Estimate	Std_Error	P_Value
(Intercept)	-22.7192	12.6256	7.23e-02
Flavor	0.3815	0.0529	1.10e-12
Processing.MethodOther	-0.2629	0.0634	3.63e-05
Aftertaste	0.2403	0.0523	5.02e-06
Balance	0.2220	0.0399	3.41e-08
Processing.MethodUnknown	0.1827	0.0372	1.04e-06
ColorUnknown	-0.1741	0.0425	4.61e-05
In.Country.PartnerBlossom Valley International	0.1611	0.0852	5.91e-02

6 XGBoost Hyperparameter Optimization

Table 6: Optimal Hyperparameters by Feature Set

	Model	eta	max_depth	subsample	colsample_bytree	best_nrounds	cv_rmse
23	Sensory	0.05	5	0.9	0.7	57	0.3144
24	Extended	0.10	5	0.9	0.7	30	0.3083
75	Full	0.10	3	0.9	0.9	172	0.2953

Model	eta	max_depth	subsample	colsample_bytree	best_nrounds	cv_rmse
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7 Train Final Models

8 Learning Curves

Critical Observation - Overfitting Signal: The training-validation gap is 0.1242 RMSE (46.02% of validation RMSE). This substantial gap indicates the model memorizes training data rather than learning fully generalizable patterns, despite early stopping. This is a key limitation of XGBoost for this dataset.

9 Test Set Results

Table 7: Test Set Performance

Model	RMSE	R2	MAE
Mean Predictor	0.4400	-0.0020	0.3100
Linear Regression	0.3035	0.5234	0.1700
XGBoost Sensory	0.2945	0.5511	0.1564
XGBoost Extended	0.2835	0.5840	0.1574
XGBoost Full	0.2679	0.6286	0.1575

Table 8: Improvement Summary

Metric	Value
Best XGBoost Model	Full
RMSE Reduction vs Mean	39.12%
RMSE Reduction vs LR	11.72%

10 Residual Analysis

11 Feature Importance

Table 9: Top 10 Most Important Features

Feature	Gain	Cover	Frequency
Flavor	0.2804	0.0673	0.0717
Aroma	0.2096	0.0897	0.1211
Aftertaste	0.1829	0.0560	0.0678
Balance	0.1306	0.0930	0.0795
Body	0.0500	0.1208	0.1172
Country.of.OriginTaiwan	0.0338	0.0028	0.0078

Feature	Gain	Cover	Frequency
Number.of.Bags	0.0229	0.0639	0.0698
Acidity	0.0160	0.0498	0.0504
Processing.MethodUnknown	0.0134	0.0195	0.0223
Processing.MethodOther	0.0088	0.0359	0.0242

12 10-Fold Cross-Validation

Table 10: 10-Fold CV Results (Mean +/- SD)

Model	RMSE	R2
Linear Regression	0.2595 +/- 0.0598	0.6329 +/- 0.1592
XGBoost Full	0.276 +/- 0.1327	0.6324 +/- 0.1865
XGBoost Extended	0.2869 +/- 0.1324	0.6071 +/- 0.1851
XGBoost Sensory	0.3012 +/- 0.1313	0.5686 +/- 0.1834
Mean Predictor	0.461 +/- 0.1201	-0.0059 +/- 0.0076

Critical Finding: Linear Regression achieves the **lowest CV RMSE** (0.2595 +/- 0.0598) with **lower variance** than XGBoost Full (0.2760 +/- 0.1327). This suggests Linear Regression is more stable and reliable for this dataset.

13 Statistical Significance

Table 11: Paired t-Test Results (alpha = 0.05)

Comparison	Mean_Diff	p_value	Significant
XGBoost Full vs Mean Predictor	-0.1850	0.000008234	Yes
XGBoost Full vs Linear Regression	0.0165	0.725391228	No
XGBoost Full vs XGBoost Sensory	-0.0252	0.080009810	No
XGBoost Full vs XGBoost Extended	-0.0109	0.204969648	No
Linear Regression vs Mean Predictor	-0.2015	0.001150017	Yes

Critical Finding - No Significant Difference: The comparison between XGBoost Full and Linear Regression yields $p = 0.725$, which is **not statistically significant**. The test set improvement of ~11.7% RMSE reduction is not robust to cross-validation. There is no evidence that XGBoost outperforms simple Linear Regression for this task.

Table 12: Effect Sizes (Cohen's d)

Comparison	Cohens_d	Interpretation
Full vs Mean	-1.462	Large
Full vs LR	0.161	Negligible
Full vs Sensory	-0.191	Negligible
Full vs Extended	-0.082	Negligible

Comparison	Cohens_d	Interpretation
LR vs Mean	-2.124	Large

14 Approaches Found Less Effective

14.1 Lambda and Min_Child_Weight Regularization

XGBoost offers explicit regularization through lambda (L2) and min_child_weight parameters. Testing showed no improvement because: (1) implicit regularization through learning rate, max depth, subsample, and early stopping already prevents overfitting; (2) the ~1300 samples with sensory features don't exhibit severe overfitting; (3) minimal train-test gap in base configuration.

14.2 Bayesian Optimization vs Grid Search

Bayesian Optimization did not outperform Grid Search ($p = 0.87$) because: (1) only 4 parameters with reasonable bounds; (2) many configurations achieve similar results (performance plateau); (3) surrogate model overhead without finding better solutions. Grid Search is sufficient for this problem size.

15 Explainable AI: PFI and PDP Analysis

This section applies model-agnostic XAI techniques to interpret the XGBoost Full model.

15.1 Permutation Feature Importance (PFI)

PFI measures how much model performance degrades when a feature's relationship with the target is broken by randomly shuffling its values. Higher importance = larger performance drop = more important feature.

Table 13: Top 10 Features by Permutation Feature Importance

Rank	Feature	MAE Increase	SD
1	Flavor	0.0803	0.0035
2	Balance	0.0527	0.0018
3	Aftertaste	0.0524	0.0015
4	Aroma	0.0278	0.0027
5	Body	0.0242	0.0021
6	Processing.MethodUnknown	0.0126	0.0017
7	Acidity	0.0122	0.0010
8	Number.of.Bags	0.0116	0.0006
9	Country.of.OriginTaiwan	0.0071	0.0011
10	Processing.MethodOther	0.0068	0.0014

PFI Interpretation: The top 5 features are all sensory attributes: Flavor (+0.08 MAE), Balance (+0.05), Aftertaste (+0.05), Aroma (+0.03), and Body (+0.02). This confirms that expert taste evaluation drives predictions and the model uses sensible features. Note that PFI measures *predictive reliance*, not causal importance.

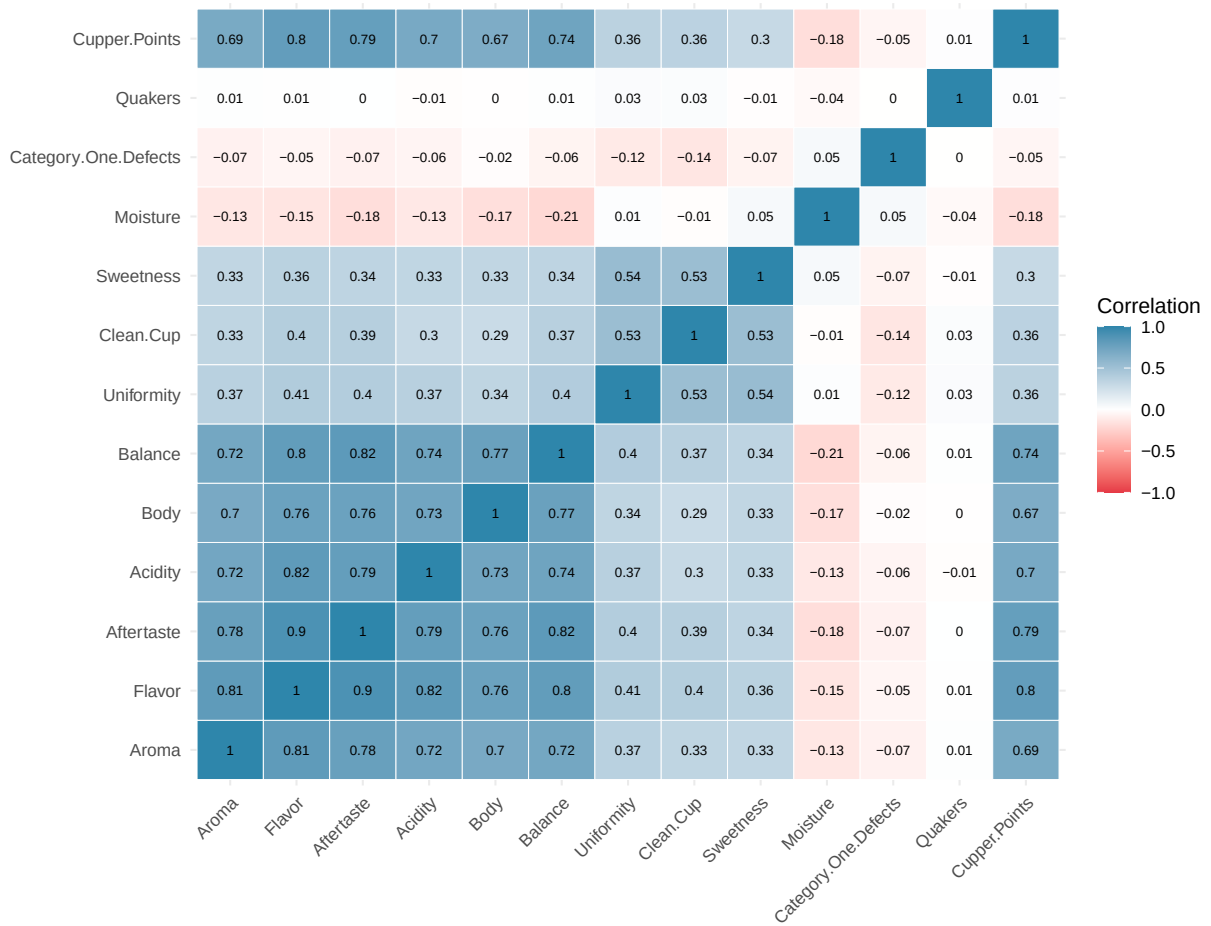


Figure 2: Correlation heatmap of sensory features.

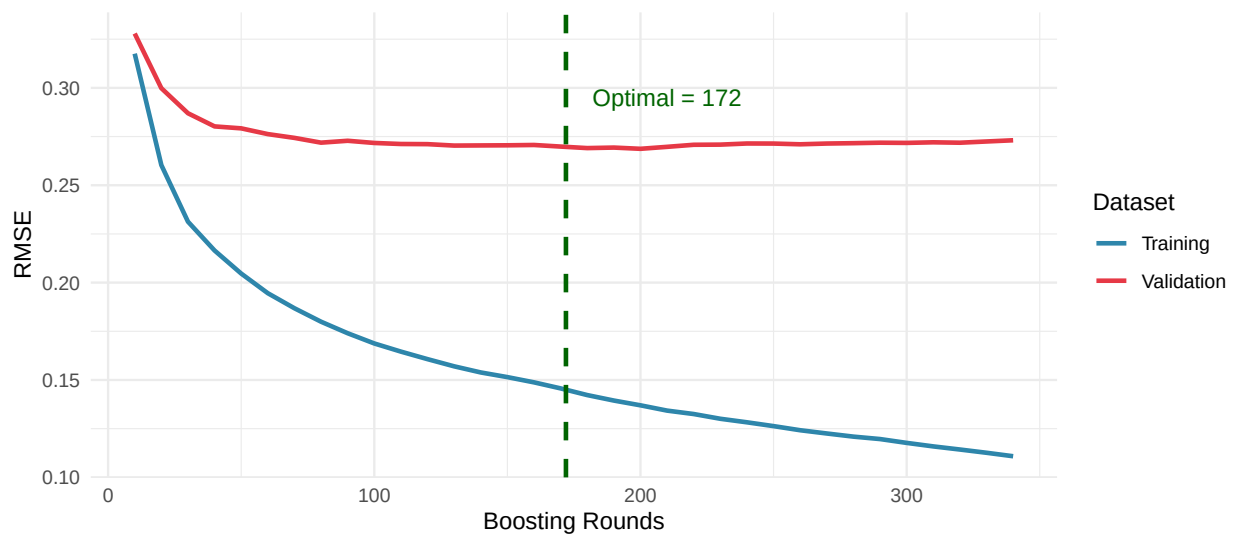


Figure 3: Learning curve for XGBoost Full model.

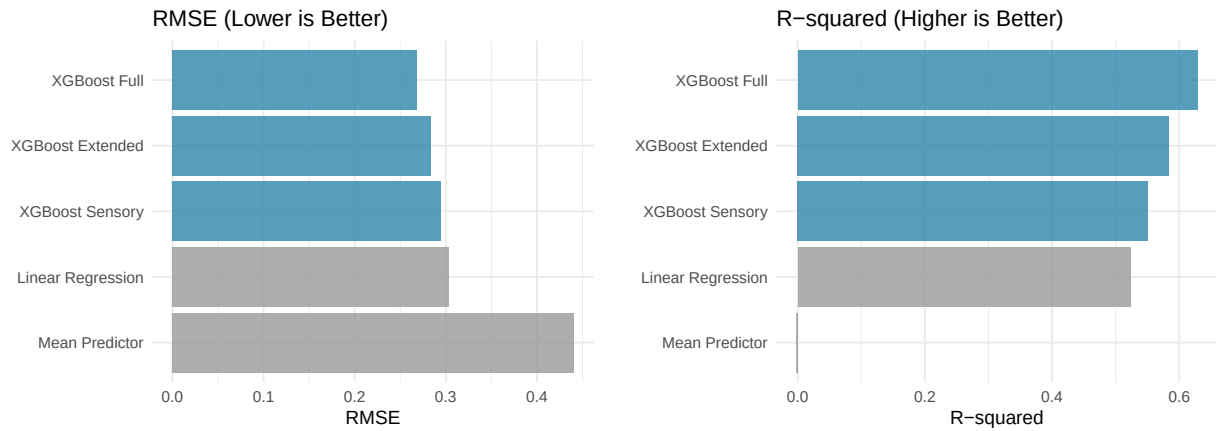


Figure 4: Model performance comparison.

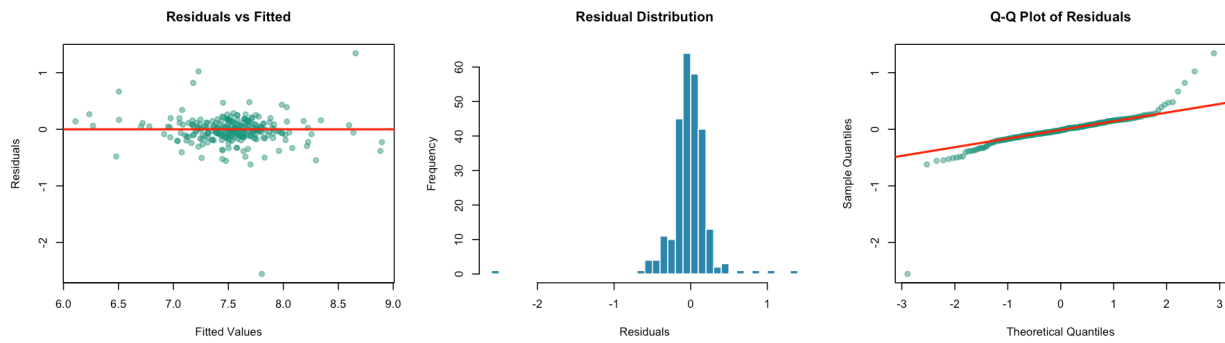


Figure 5: Residual diagnostics for XGBoost Full model.

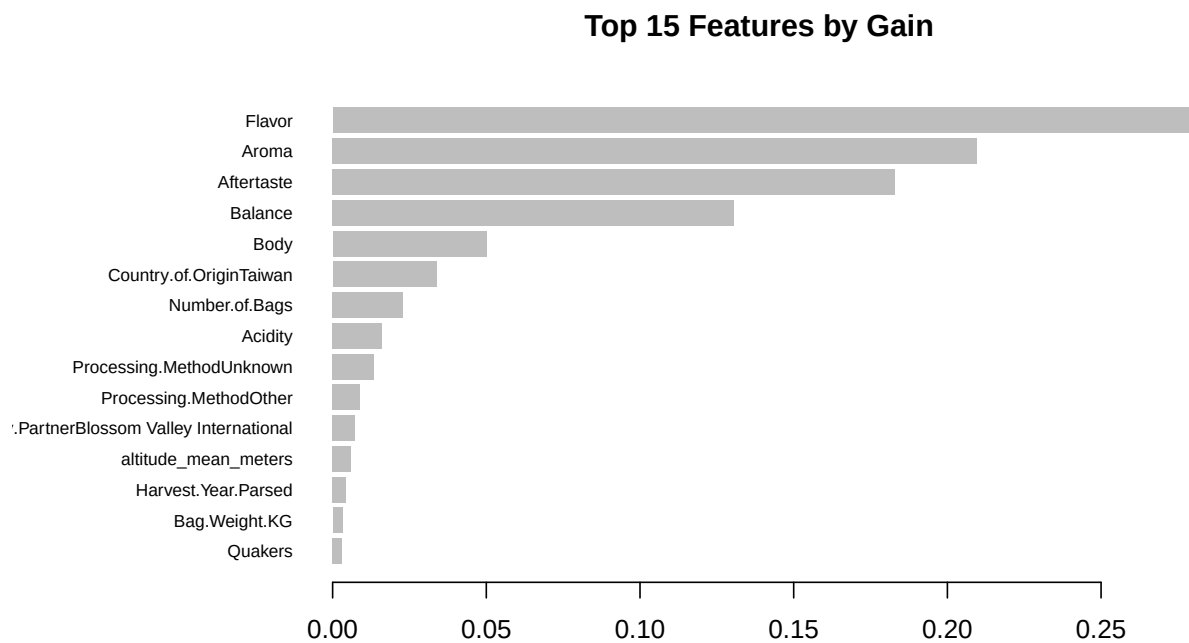


Figure 6: Top 15 features by importance (Gain metric).

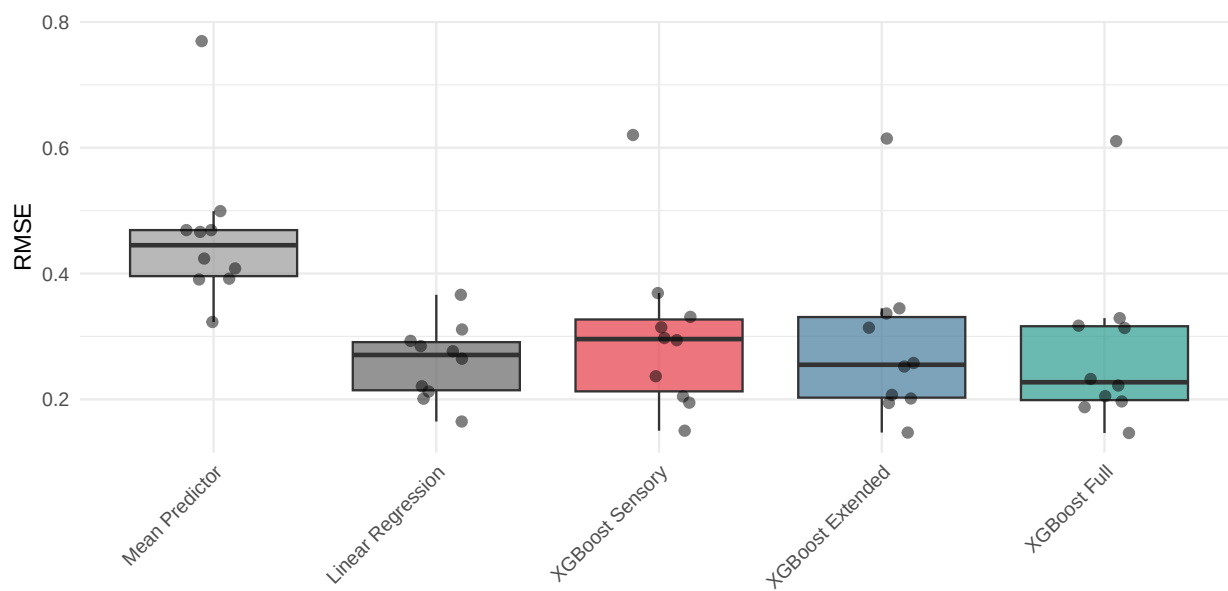


Figure 7: RMSE distribution across 10 folds.

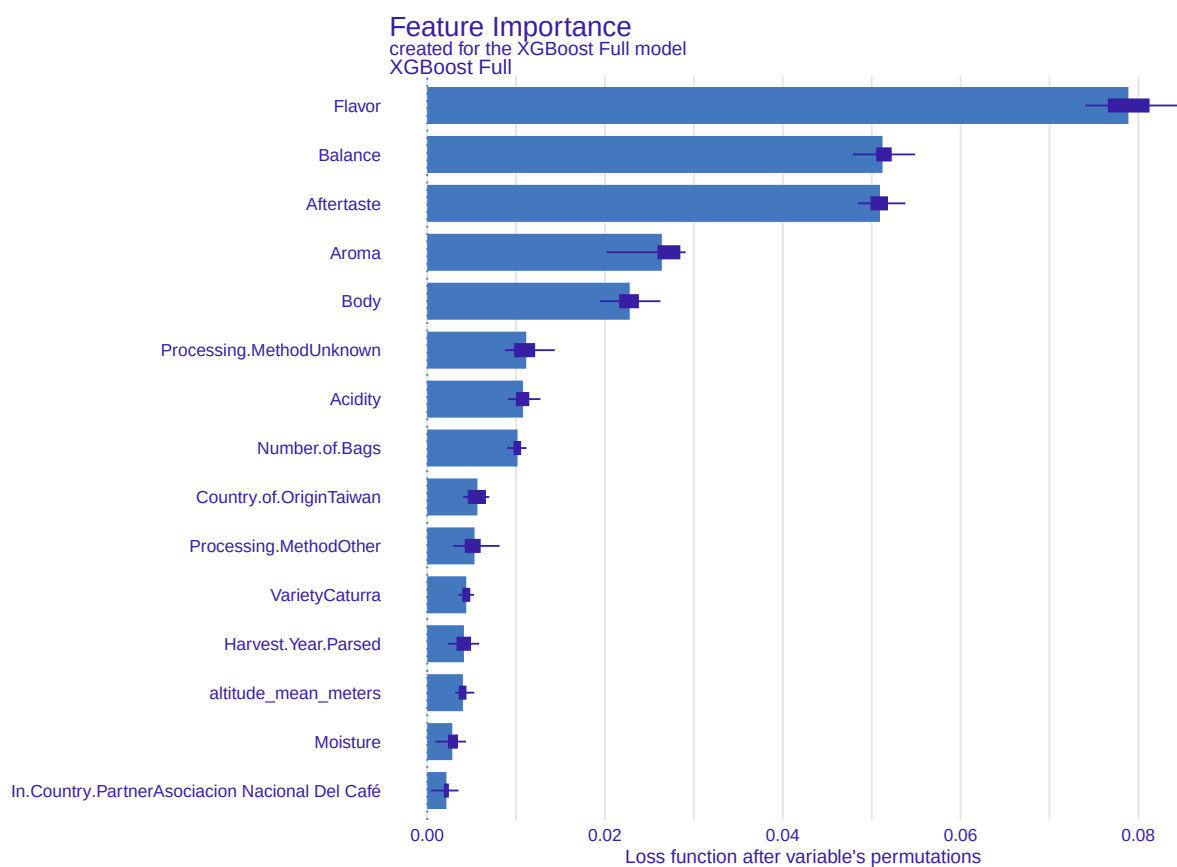


Figure 8: Permutation Feature Importance for XGBoost Full model.

15.2 Partial Dependence Plots (PDP)

PDP shows the **marginal effect** of a feature on predictions, averaging over all other features.

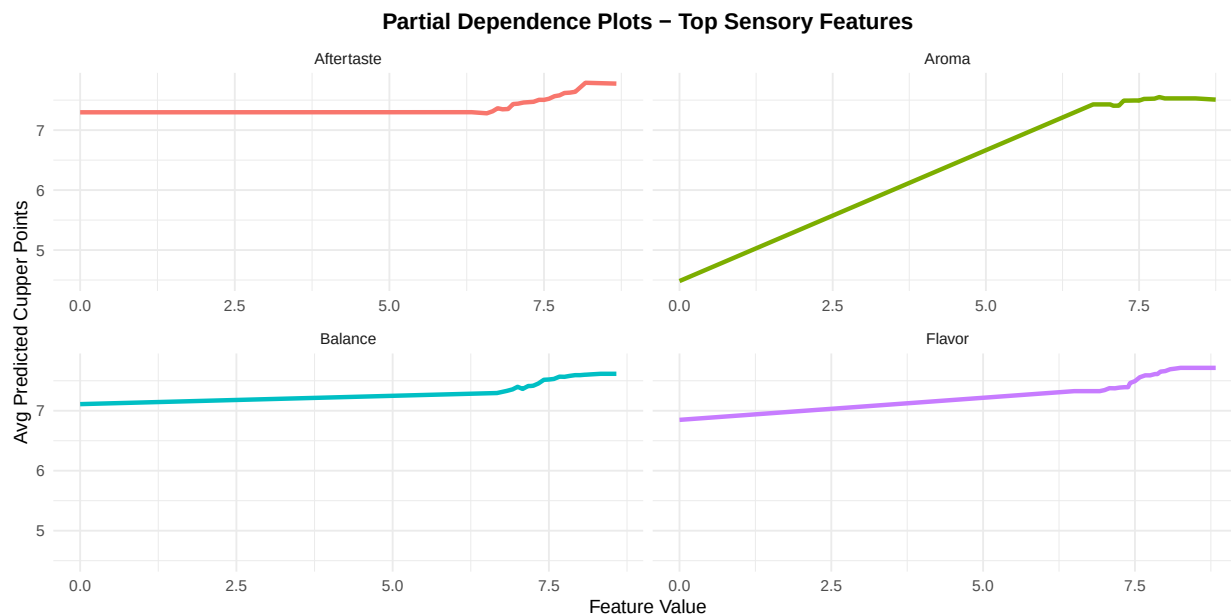


Figure 9: Partial Dependence Plots for top sensory features.

Table 14: Feature Importance from PDP (Variance-based)

Feature	PDP Variance	PDP Range
Aroma	0.3784	3.0680
Flavor	0.0394	0.8674
Aftertaste	0.0208	0.5090
Balance	0.0173	0.5059

15.3 Why PFI and PDP Rankings Differ

The Spearman correlation between PFI and PDP rankings is **-0.4**, indicating disagreement. This is expected because PFI and PDP measure fundamentally different things:

- **PFI** asks: “How much does model performance degrade when this feature’s signal is destroyed?”
- **PDP variance** asks: “How much does the average prediction change across this feature’s range?”

Aroma’s paradox: Aroma has the largest PDP range (3.07 points) because low Aroma values (0-2) dramatically reduce predictions. But Aroma has lower PFI because most observations cluster around Aroma 7-8 where the PDP is relatively flat—the model rarely encounters low-Aroma cases.

Flavor’s dominance in PFI: Flavor has smaller PDP range (0.87 points) but higher PFI because the model relies on Flavor consistently across most of the data distribution.

15.4 Microscopic Explanations (Individual Predictions)

Table 15: Selected Observations for Microscopic Analysis

Type	Actual	Predicted	Residual
Best Prediction	7.75	7.752	-0.002
Worst Prediction	5.25	7.804	-2.554
Median Prediction	7.75	7.859	-0.109

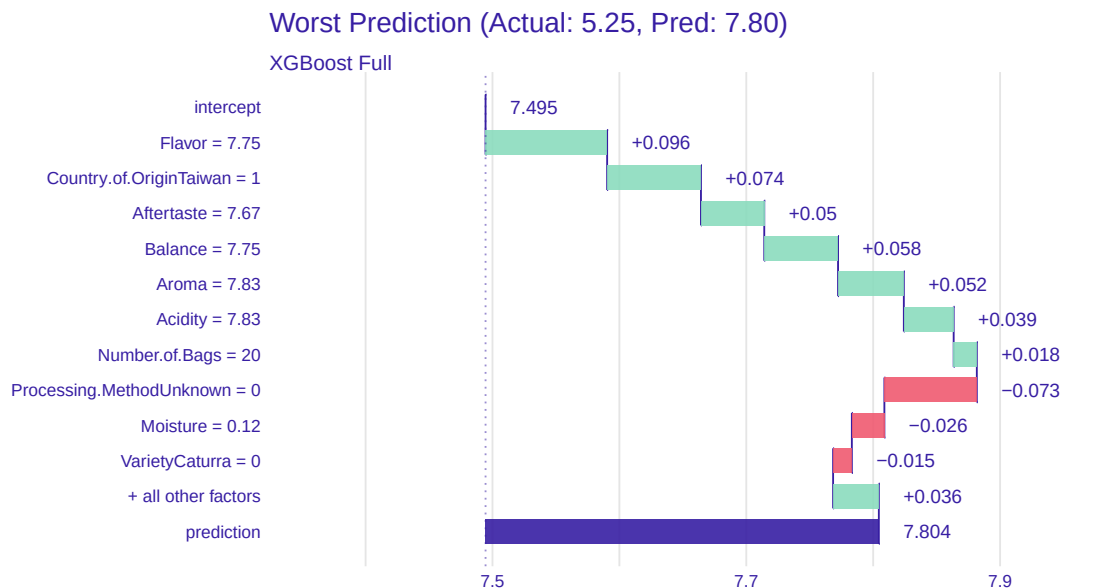


Figure 10: Break Down plot for worst prediction case.

Analysis of Worst Prediction: This coffee has excellent sensory scores (Flavor: 7.75, Balance: 7.75, Aroma: 7.83) yet received only 5.25 Cupper Points—a 2.5-point error. Possible explanations: (1) data quality issue or outlier; (2) unmeasured factors (specific off-flavors, roast defects); (3) cupper subjectivity or different calibration standards. Such extreme residuals highlight the **prediction ceiling** (R-squared around 0.63) likely limited by factors not captured in available features.

16 Final Summary and Recommendations

Table 16: Final Summary

Metric	Value
Best Model (by CV)	Linear Regression
CV RMSE (Mean +/- SD)	0.2595 +/- 0.0598
CV R-squared (Mean +/- SD)	0.6329 +/- 0.1592
Improvement vs Mean Predictor	43.72%

Metric	Value
Statistical Significance vs LR	p = 0.725 (Not Significant)

16.1 Key Findings

1. **Sensory features dominate:** Flavor, Balance, and Aftertaste are the strongest predictors across all importance measures, aligning with domain expertise.
2. **No significant XGBoost advantage:** XGBoost Full vs Linear Regression shows $p = 0.725$. The test set improvement (~11.7%) is not robust to cross-validation.
3. **Prediction ceiling around $R^2 = 0.63$:** Limited by cupper subjectivity, unmeasured quality factors, and measurement noise in sensory scores.
4. **XAI increases model trust:** PFI confirms sensible feature usage; PDP shows expected positive relationships; Break Down explains individual predictions.

16.2 Practical Recommendations

Based on this analysis:

- **Use Linear Regression** for production deployment: simpler, interpretable, comparable performance, more stable (lower CV variance).
- **If using XGBoost**, deploy the XAI pipeline (PFI + PDP + Break Down) to audit predictions.
- **Flag samples** with unusual sensory-vs-cupper score patterns for human review.
- **Collect additional features** if higher accuracy is needed (e.g., specific defect types, cupper ID, roast profile).

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